Example 8.4.2

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The rate expressions for reactions (1) and (2) are shown in equations (3) and (4). The Arrhenius parameters for these reactions are $k_{0,1}$ = 1.83 x 10^{12} L mol^{-1} h^{-1}, E_1 = 18.0 kcal mol^{-1}, $k_{0,2}$ = 5.08 x 10^{13} L mol^{-1} h^{-1}, and E_2 = 20.5 kcal mol^{-1}, Reagent D is the desired product while reagent U is undesired. The heats of reactions (1) and (2) are -9000 and -7800 cal mol^{-1}, respectively, and solutions of A and B have a heat capacity of 863 cal L^{-1} K^{-1}.

A 2 M solution of A and a 0.5 M solution of B, both at 40 °C, are going to be used to produce D in an adiabatic semi-batch reactor operating at 1 atm. The reactor will be charged with 2000 L of the A solution, and 8000 L of the B solution will be added at a constant rate over a period of time. If the total reaction time is always 8 h, compare the overall conversion of B, the selectivity for D over U (final moles of D per final moles of U), and the yield of D from B (final moles of D per total moles of B added to the reactor) if the solution of B is added over the first 1, 3, 5, or 7 hours of operation.

$$A + B \rightarrow D \tag{1}$$

$$2B \rightarrow U \tag{2}$$

$$r_1 = k_1 C_A C_B \tag{3}$$

$$r_2 = k_2 C_B^2 \tag{4}$$